

GED Math Practice Test 13

PART A — QUESTIONS

GED-MATH13-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A food pantry packed 3 crates with 24 cans in each crate and 2 crates with 18 cans in each crate. If 30 cans were given out before noon, how many cans remained?

- A) 48
- B) 66
- C) 78
- D) 108

GED-MATH13-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of $3x^2 - 2x$ when $x = 4$?

- A) 32
- B) 36
- C) 40
- D) 48

GED-MATH13-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A container is $\frac{5}{8}$ full. Then $\frac{1}{4}$ of the container is added, and $\frac{3}{8}$ of the container is used. What fraction of the container is full after these changes?

- A) $\frac{1}{4}$
- B) $\frac{1}{2}$
- C) $\frac{3}{4}$
- D) $\frac{7}{8}$

GED-MATH13-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to $-5(2y + 1) + 3(y - 4)$?

- A) $-7y - 17$
- B) $-7y + 7$

- C) $-13y - 17$
D) $13y - 7$

GED-MATH13-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A training group has 15 new employees and 45 returning employees. What percent of the group is made up of new employees?

- A) 15%
B) 20%
C) 25%
D) 33%

GED-MATH13-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $H = 4(t - 2)^2 + 10$ gives the height H , in inches, of a model after t seconds. If $H = 46$, what is one possible value of t ?

- A) 2
B) 3
C) 5
D) 11

GED-MATH13-007

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A clinic orders 12 boxes of bandage packets. Each box contains 25 packets, and each packet weighs 40 grams. What is the total weight of the packets in kilograms?

Type your answer as an integer.

GED-MATH13-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $A = P(1 + rt)$ gives the amount A after simple interest. To solve for r , first divide both sides by P , subtract $[D1]$, and then divide by $[D2]$.

Dropdowns:

- D1) 1 | t | P
D2) 1 | t | P

GED-MATH13-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A carpenter has 6 boards that are 8.5 feet long and 4 boards that are 6.25 feet long. If 7.5 feet of wood is lost to cuts and waste, how many feet of usable wood remain?

A) 61.0 feet

B) 68.5 feet

C) 76.0 feet

D) 83.5 feet

GED-MATH13-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which statement describes all possible values of x if $2(3x - 5) + 4 < 30$?

A) $x < 6$

B) $x > 6$

C) $x < 8$

D) $x > 8$

GED-MATH13-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A table shows pages read during three study sessions.

Session: A, B, C

Pages read: 90, 132, 108

Time spent: 1.5 hr, 2 hr, 1.8 hr

Which session had the greatest number of pages read per hour?

A) Session A

B) Session B

C) Session C

D) Sessions A and C were tied

GED-MATH13-012

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which expressions are equivalent to $2(4x - 3) - 5(x - 2)$?

- A) $3x + 4$
- B) $8x - 6 - 5x + 10$
- C) $13x - 16$
- D) $2(4x - 3) - 5x - 2$
- E) $3(x + 4)$

GED-MATH13-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A volunteer program recorded how many people signed up for each task.

Task: Packing, Delivery, Inventory

Morning volunteers: 12, 8, 10

Afternoon volunteers: 15, 9, 6

If one afternoon volunteer is selected at random, what is the probability that the volunteer signed up for delivery?

- A) $1/5$
- B) $3/10$
- C) $2/5$
- D) $9/50$

GED-MATH13-014

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A formula for an event score is $E = 0.5r + 2s$, where r is the number of regular points and s is the number of special points. If $E = 47$ and $r = 30$, what is s ?

Type your answer as an integer.

GED-MATH13-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

An L-shaped floor can be described as a 20-foot by 14-foot rectangle with a 6-foot by 5-foot corner removed. What is the area of the L-shaped floor?

- A) 220 square feet
- B) 250 square feet
- C) 280 square feet
- D) 310 square feet

GED-MATH13-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows part of a linear relationship.

x: -4, 0, 6, 10

y: 24, 14, -1, -11

Which equation represents the relationship?

A) $y = -2x + 16$

B) $y = -2.5x + 14$

C) $y = 2.5x + 34$

D) $y = -4x + 8$

GED-MATH13-017

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A store compares two boxes of printer paper.

Box: A, B

Sheets: 1,500, 2,400

Price: \$12.75, \$18.72

Which statements are true?

A) Box A costs \$0.0085 per sheet.

B) Box B costs \$0.0078 per sheet.

C) Box A has the lower unit price.

D) Box B costs \$0.0092 per sheet.

E) The two boxes have the same unit price.

GED-MATH13-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which inequality represents “twice the sum of a number and 5 is at least the difference of 60 and the number”?

A) $2x + 5 \geq 60 - x$

B) $2(x + 5) \geq 60 - x$

C) $2(x - 5) \geq 60 + x$

D) $2(x + 5) \leq 60 - x$

GED-MATH13-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A customer paid \$74.20 for an item after 6% sales tax was added. What was the price of the item before tax?

- A) \$68.00
- B) \$70.00
- C) \$78.65
- D) \$80.20

GED-MATH13-020

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for x: $\frac{2}{3}x - 4 = 18$.

Type your answer as an integer.

GED-MATH13-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular dog pen is 28 feet long and 16 feet wide. Fencing is needed around the entire pen except for a 4-foot gate opening. How many feet of fencing are needed?

- A) 72 feet
- B) 80 feet
- C) 84 feet
- D) 88 feet

GED-MATH13-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A line passes through the points $(-3, 18)$ and $(9, -6)$. What is the rate of change of the line?

- A) -3
- B) -2
- C) 2
- D) 3

GED-MATH13-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A community group has a \$900 event budget. It spends \$240 on supplies, \$315 on the venue, and \$126 on advertising. What percent of the budget remains?

- A) 18.7%
- B) 21.9%
- C) 24.3%
- D) 31.8%

GED-MATH13-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Function A is represented by the table.

x: -2, 2, 8

y: 18, 14, 8

Function B decreases by 6 when x increases by 3. Which statement is true?

- A) Function A decreases faster than Function B.
- B) Function B decreases faster than Function A.
- C) Both functions have the same rate of change.
- D) Function A has a positive rate of change.

GED-MATH13-025

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A store has 18 items that cost \$12 each, 32 items that cost \$9.50 each, and 10 items that cost \$15 each. What is the average cost per item?

Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH13-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A class bought 48 notebooks. Single-subject notebooks cost \$2.40 each, and three-subject notebooks cost \$4.80 each. The total cost was \$158.40. How many three-subject notebooks were bought?

- A) 12
- B) 16
- C) 18
- D) 24

GED-MATH13-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

On a scale drawing, 3 centimeters represents 8 feet. A rectangular patio is 7.2 centimeters by 4.5 centimeters on the drawing. What is the actual area of the patio?

- A) 86.4 square feet
- B) 115.2 square feet
- C) 192.0 square feet
- D) 230.4 square feet

GED-MATH13-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation $5x + 4y = 3x + 28$ can be rewritten as $y = [D1]x + [D2]$.

Dropdowns:

D1) $-1/2$ | $1/2$ | 2

D2) 7 | 28 | -7

GED-MATH13-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A quality-control bin contains 21 working sensors and 4 defective sensors. Two sensors are selected at random without replacement. What is the probability that both selected sensors are working?

- A) $7/10$
- B) $21/25$
- C) $20/24$
- D) $441/625$

GED-MATH13-030

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

The formula $F = 9/5C + 32$ converts Celsius C to Fahrenheit F . Which equations correctly solve for C ?

- A) $C = 5(F - 32)/9$
- B) $C = 9(F - 32)/5$
- C) $C = (F - 32)/(9/5)$
- D) $C = F - 32/(9/5)$
- E) $C = 5F/9 + 32$

GED-MATH13-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A liquid has a density of 1.2 grams per milliliter. A 750-milliliter bottle is filled to 80% of its capacity. What is the mass of the liquid in the bottle?

- A) 600 grams
- B) 720 grams
- C) 900 grams
- D) 1,125 grams

GED-MATH13-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for z : $5(2z + 1) - 3(4 - z) = 97$.

- A) 6
- B) 8
- C) 10
- D) 12

GED-MATH13-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cylindrical tank has a radius of 4 feet and a height of 15 feet. It is filled to 60% of its full volume. Using $\pi \approx 3.14$, how many cubic feet of liquid are in the tank?

- A) 301.44 ft³
- B) 376.80 ft³
- C) 452.16 ft³
- D) 753.60 ft³

GED-MATH13-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The height h , in feet, of a thrown ball after t seconds is modeled by $h = -16t^2 + 48t + 5$. What is the height of the ball after 2 seconds?

- A) 21 feet
- B) 37 feet
- C) 69 feet
- D) 101 feet

GED-MATH13-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A sample starts with 640 bacteria and half of the bacteria remain after each hour. Which value gives the number of bacteria remaining after 3 hours?

- A) 80
- B) 160
- C) 320
- D) 640

GED-MATH13-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A medicine label says 250 milligrams are contained in 5 milliliters. How many milliliters are needed for a 600-milligram dose?

- A) 10 mL
- B) 12 mL
- C) 15 mL
- D) 18 mL

GED-MATH13-037

Section: Math

Type: drag_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A) $12x + 7y = 95$
- B) $x + y = 95$
- C) $x = y + 7$
- D) $y = x + 7$

Targets:

T1) Adult tickets cost \$12 each, child tickets cost \$7 each, and the total cost is \$95

T2) The total number of adult and child tickets is 95

T3) The number of adult tickets is 7 more than the number of child tickets

GED-MATH13-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Printer A prints 84 pages in 7 minutes. Printer B prints 90 pages in 10 minutes. If both printers run at these rates for 15 minutes, how many pages will they print together?

- A) 270
- B) 300
- C) 315
- D) 360

GED-MATH13-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

An elevator can carry at most 1,200 pounds. Equipment in the elevator weighs 180 pounds. Each crate weighs 65 pounds. What is the greatest number of crates that can be added?

A) 14

B) 15

C) 16

D) 18

GED-MATH13-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A website has 1,600 members and grows by 7% each quarter. Which value is closest to the number of members after 2 quarters?

A) 1,712

B) 1,832

C) 1,920

D) 2,289

GED-MATH13-041

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A teacher bought supplies for a class.

Item: Binders, Notebooks, Pen packs

Quantity: 6, 12, 4

Unit price: \$3.25, \$1.80, \$2.75

Which statements are true?

A) The total cost was \$52.10.

B) The notebooks cost more in total than the binders.

C) The pen packs cost \$10.00 in total.

D) The average cost per item type was exactly \$17.00.

E) The binders cost \$21.60 in total.

GED-MATH13-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function $f(x) = 0.5x^2 - 3x + 8$, what is $f(6)$?

- A) 2
- B) 8
- C) 18
- D) 44

GED-MATH13-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows a relationship.

x: -1, 0, 1, 2

y: 7, 4, 3, 4

Which equation represents the relationship?

- A) $y = x^2 - 2x + 4$
- B) $y = 2x + 4$
- C) $y = -3x + 4$
- D) $y = x^2 + 4$

GED-MATH13-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which expression represents 5 less than the quotient of 18 and the sum of a number x and 2?

- A) $18/(x + 2) - 5$
- B) $18/x + 2 - 5$
- C) $5 - 18/(x + 2)$
- D) $(18/x + 2)/5$

GED-MATH13-045

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $V = 1/3Bh$ gives the volume V of a pyramid, where B is the area of the base and h is the height. If $V = 180$ and $B = 45$, what is h ?

Type your answer as an integer.

GED-MATH13-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A floor plan includes a rectangular room that is 30 feet by 22 feet and a semicircular alcove attached to one side. The semicircle has a diameter of 10 feet. Using $\pi \approx 3.14$, what is the total floor area?

- A) 660.00 square feet
- B) 699.25 square feet
- C) 817.00 square feet
- D) 974.00 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH13-001 — Correct: C — Correct Answer: 78 — Explanation: The pantry packed $3(24) + 2(18) = 72 + 36 = 108$ cans. After 30 cans were given out, $108 - 30 = 78$ remained.

GED-MATH13-002 — Correct: C — Correct Answer: 40 — Explanation: Substitute $x = 4$: $3(4^2) - 2(4) = 48 - 8 = 40$.

GED-MATH13-003 — Correct: B — Correct Answer: $\frac{1}{2}$ — Explanation: $\frac{5}{8} + \frac{1}{4} - \frac{3}{8} = \frac{5}{8} + \frac{2}{8} - \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$.

GED-MATH13-004 — Correct: A — Correct Answer: $-7y - 17$ — Explanation: $-5(2y + 1) + 3(y - 4) = -10y - 5 + 3y - 12 = -7y - 17$.

GED-MATH13-005 — Correct: C — Correct Answer: 25% — Explanation: There are 60 employees total, and 15 are new. Then $15/60 = 0.25$, or 25%.

GED-MATH13-006 — Correct: C — Correct Answer: 5 — Explanation: $46 = 4(t - 2)^2 + 10$, so $36 = 4(t - 2)^2$ and $9 = (t - 2)^2$. One solution is $t - 2 = 3$, so $t = 5$.

GED-MATH13-007 — Correct: 12 — Tolerance: exact — Correct Answer: 12 — Explanation: The packets weigh $12 \times 25 \times 40 = 12,000$ grams, which is 12 kilograms.

GED-MATH13-008 — Correct: $D1=1$; $D2=t$ — Correct Answer: $D1=1$; $D2=t$ — Explanation: Divide by P to get $A/P = 1 + rt$. Subtract 1 to get $A/P - 1 = rt$, then divide by t .

GED-MATH13-009 — Correct: B — Correct Answer: 68.5 feet — Explanation: Total board length is $6(8.5) + 4(6.25) = 51 + 25 = 76$ feet. Usable wood is $76 - 7.5 = 68.5$ feet.

GED-MATH13-010 — Correct: A — Correct Answer: $x < 6$ — Explanation: $2(3x - 5) + 4 < 30$ simplifies to $6x - 6 < 30$, so $6x < 36$ and $x < 6$.

GED-MATH13-011 — Correct: B — Correct Answer: Session B — Explanation: Session A: $90 \div 1.5 = 60$ pages per hour. Session B: $132 \div 2 = 66$. Session C: $108 \div 1.8 = 60$. Session B had the greatest rate.

GED-MATH13-012 — Correct: A,B — Correct Answer: $3x + 4 \mid 8x - 6 - 5x + 10$ — Explanation: $2(4x - 3) - 5(x - 2) = 8x - 6 - 5x + 10 = 3x + 4$.

GED-MATH13-013 — Correct: B — Correct Answer: $\frac{3}{10}$ — Explanation: There are $15 + 9 + 6 = 30$ afternoon volunteers, and 9 signed up for delivery. The probability is $9/30 = \frac{3}{10}$.

GED-MATH13-014 — Correct: 16 — Tolerance: exact — Correct Answer: 16 — Explanation: Substitute $E = 47$ and $r = 30$: $47 = 0.5(30) + 2s$, so $47 = 15 + 2s$. Then $32 = 2s$ and $s = 16$.

GED-MATH13-015 — Correct: B — Correct Answer: 250 square feet — Explanation: The large rectangle has area $20 \times 14 = 280$. The removed corner has area $6 \times 5 = 30$. The L-shaped area is $280 - 30 = 250$.

GED-MATH13-016 — Correct: B — Correct Answer: $y = -2.5x + 14$ — Explanation: From $x = -4$ to $x = 0$, y changes from 24 to 14, so the rate is $-10/4 = -2.5$. When $x = 0$, $y = 14$.

GED-MATH13-017 — Correct: A,B — Correct Answer: Box A costs \$0.0085 per sheet. | Box B costs \$0.0078 per sheet. — Explanation: Box A: $12.75 \div 1,500 = 0.0085$. Box B: $18.72 \div 2,400 = 0.0078$.

GED-MATH13-018 — Correct: B — Correct Answer: $2(x + 5) \geq 60 - x$ — Explanation: The sum of a number and 5 is $x + 5$, twice that is $2(x + 5)$, and “at least” means $\geq$.

GED-MATH13-019 — Correct: B — Correct Answer: \$70.00 — Explanation: The total after tax is 106% of the original price. So the original price is $74.20 \div 1.06 = 70.00$.

GED-MATH13-020 — Correct: 33 — Tolerance: exact — Correct Answer: 33 — Explanation: Add 4 to both sides to get $2/3x = 22$. Multiply by $3/2$ to get $x = 33$.

GED-MATH13-021 — Correct: C — Correct Answer: 84 feet — Explanation: The full perimeter is $2(28 + 16) = 88$ feet. Subtract the 4-foot gate opening: $88 - 4 = 84$ feet.

GED-MATH13-022 — Correct: B — Correct Answer: -2 — Explanation: Rate of change = $(-6 - 18)/(9 - (-3)) = -24/12 = -2$.

GED-MATH13-023 — Correct: C — Correct Answer: 24.3% — Explanation: Total spent is $240 + 315 + 126 = 681$. The remaining budget is $900 - 681 = 219$. Then $219 \div 900 = 0.243...$, or about 24.3%.

GED-MATH13-024 — Correct: B — Correct Answer: Function B decreases faster than Function A. — Explanation: Function A has rate $(14 - 18)/(2 - (-2)) = -4/4 = -1$. Function B has rate $-6/3 = -2$, so Function B decreases faster.

GED-MATH13-025 — Correct: 11.17 — Tolerance: exact — Correct Answer: 11.17 — Explanation: Total cost is $18(12) + 32(9.50) + 10(15) = 216 + 304 + 150 = 670$. There are 60 items total, and $670 \div 60 = 11.166...$, which rounds to 11.17.

GED-MATH13-026 — Correct: C — Correct Answer: 18 — Explanation: Let t be the number of three-subject notebooks. Then $2.40(48 - t) + 4.80t = 158.40$. This gives $115.20 + 2.40t = 158.40$, so $t = 18$.

GED-MATH13-027 — Correct: D — Correct Answer: 230.4 square feet — Explanation: Since 3 centimeters represents 8 feet, 1 centimeter represents $8/3$ feet. The actual dimensions are $7.2 \times 8/3 = 19.2$ feet and $4.5 \times 8/3 = 12$ feet. The area is $19.2 \times 12 = 230.4$ square feet.

GED-MATH13-028 — Correct: $D1 = -1/2$; $D2 = 7$ — Correct Answer: $D1 = -1/2$; $D2 = 7$ — Explanation: $5x + 4y = 3x + 28$ gives $4y = -2x + 28$, so $y = -1/2x + 7$.

GED-MATH13-029 — Correct: A — Correct Answer: $7/10$ — Explanation: There are 25 sensors total. Without replacement, $P(\text{two working}) = 21/25 \times 20/24 = 420/600 = 7/10$.

GED-MATH13-030 — Correct: A,C — Correct Answer: $C = 5(F - 32)/9$ | $C = (F - 32)/(9/5)$ — Explanation: From $F = 9/5C + 32$, subtract 32 and divide by $9/5$. This is the same as multiplying by $5/9$.

GED-MATH13-031 — Correct: B — Correct Answer: 720 grams — Explanation: The bottle contains $0.80 \times 750 = 600$ milliliters. Mass = density $\times$ volume = $1.2 \times 600 = 720$ grams.

GED-MATH13-032 — Correct: B — Correct Answer: 8 — Explanation: $5(2z + 1) - 3(4 - z) = 10z + 5 - 12 + 3z = 13z - 7$. Then $13z - 7 = 97$, so $z = 8$.

GED-MATH13-033 — Correct: C — Correct Answer: 452.16 ft^3 — Explanation: Full volume is $3.14(4^2)(15) = 753.60$ cubic feet. Sixty percent of that is $0.60 \times 753.60 = 452.16$ cubic feet.

GED-MATH13-034 — Correct: B — Correct Answer: 37 feet — Explanation: $h = -16(2^2) + 48(2) + 5 = -64 + 96 + 5 = 37$ feet.

GED-MATH13-035 — Correct: A — Correct Answer: 80 — Explanation: After 3 hours, the amount is $640(0.5)^3 = 640(0.125) = 80$ bacteria.

GED-MATH13-036 — Correct: B — Correct Answer: 12 mL — Explanation: The medicine concentration is $250 \div 5 = 50$ milligrams per milliliter. For 600 milligrams, $600 \div 50 = 12$ milliliters are needed.

GED-MATH13-037 — Correct: $T1=A$; $T2=B$; $T3=C$ — Correct Answer: $T1=12x + 7y = 95$; $T2=x + y = 95$; $T3=x = y + 7$ — Explanation: Each equation matches the relationship described in its target.

GED-MATH13-038 — Correct: C — Correct Answer: 315 — Explanation: Printer A prints $84 \div 7 = 12$ pages per minute. Printer B prints $90 \div 10 = 9$ pages per minute. Together they print 21 pages per minute, so in 15 minutes they print 315 pages.

GED-MATH13-039 — Correct: B — Correct Answer: 15 — Explanation: The total weight must satisfy $180 + 65c \leq 1,200$. Then $65c \leq 1,020$, so $c \leq 15.69$. The greatest whole number of crates is 15.

GED-MATH13-040 — Correct: B — Correct Answer: 1,832 — Explanation: After 2 quarters, the number of members is $1,600(1.07)^2 = 1,831.84$, which is closest to 1,832.

GED-MATH13-041 — Correct: A,B — Correct Answer: The total cost was \$52.10. | The notebooks cost more in total than the binders. — Explanation: Binders cost $6(3.25) = \$19.50$, notebooks cost $12(1.80) = \$21.60$, and pen packs cost $4(2.75) = \$11.00$. The total is \$52.10.

GED-MATH13-042 — Correct: B — Correct Answer: 8 — Explanation: $f(6) = 0.5(6^2) - 3(6) + 8 = 18 - 18 + 8 = 8$.

GED-MATH13-043 — Correct: A — Correct Answer: $y = x^2 - 2x + 4$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $y = x^2 - 2x + 4$ gives 7, 4, 3, and 4.

GED-MATH13-044 — Correct: A — Correct Answer: $18/(x + 2) - 5$ — Explanation: The sum of x and 2 is $x + 2$. The quotient of 18 and that sum is $18/(x + 2)$, and 5 less than that is $18/(x + 2) - 5$.

GED-MATH13-045 — Correct: 12 — Tolerance: exact — Correct Answer: 12 — Explanation: $180 = \frac{1}{3}(45)h$, so $180 = 15h$ and $h = 12$.

GED-MATH13-046 — Correct: B — Correct Answer: 699.25 square feet — Explanation: The rectangular room area is $30 \times 22 = 660$ square feet. The semicircle has radius 5 feet, so its area is $\frac{1}{2}(3.14)(5^2) = 39.25$ square feet. The total area is $660 + 39.25 = 699.25$ square feet.